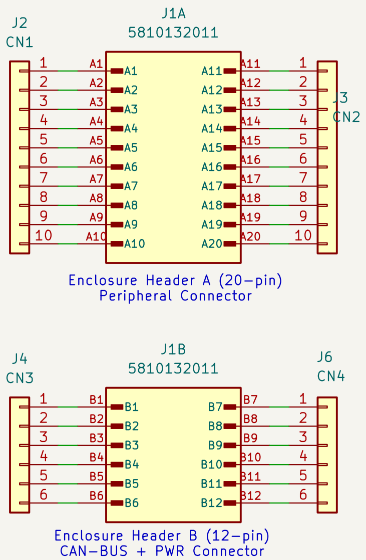
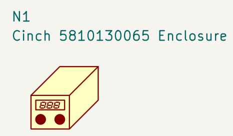
PCB/Enclosure Output Connectors





These are the main inputs/outputs from the enclosure that will connect to the wiring harness on the car.

Internally they are connected to pin headers inside the enclosure so that the pinout is completely adjustable for prototyping different pinouts and connections on the MCU

VCU Module TOP-LEVEL ARCHITECTURE

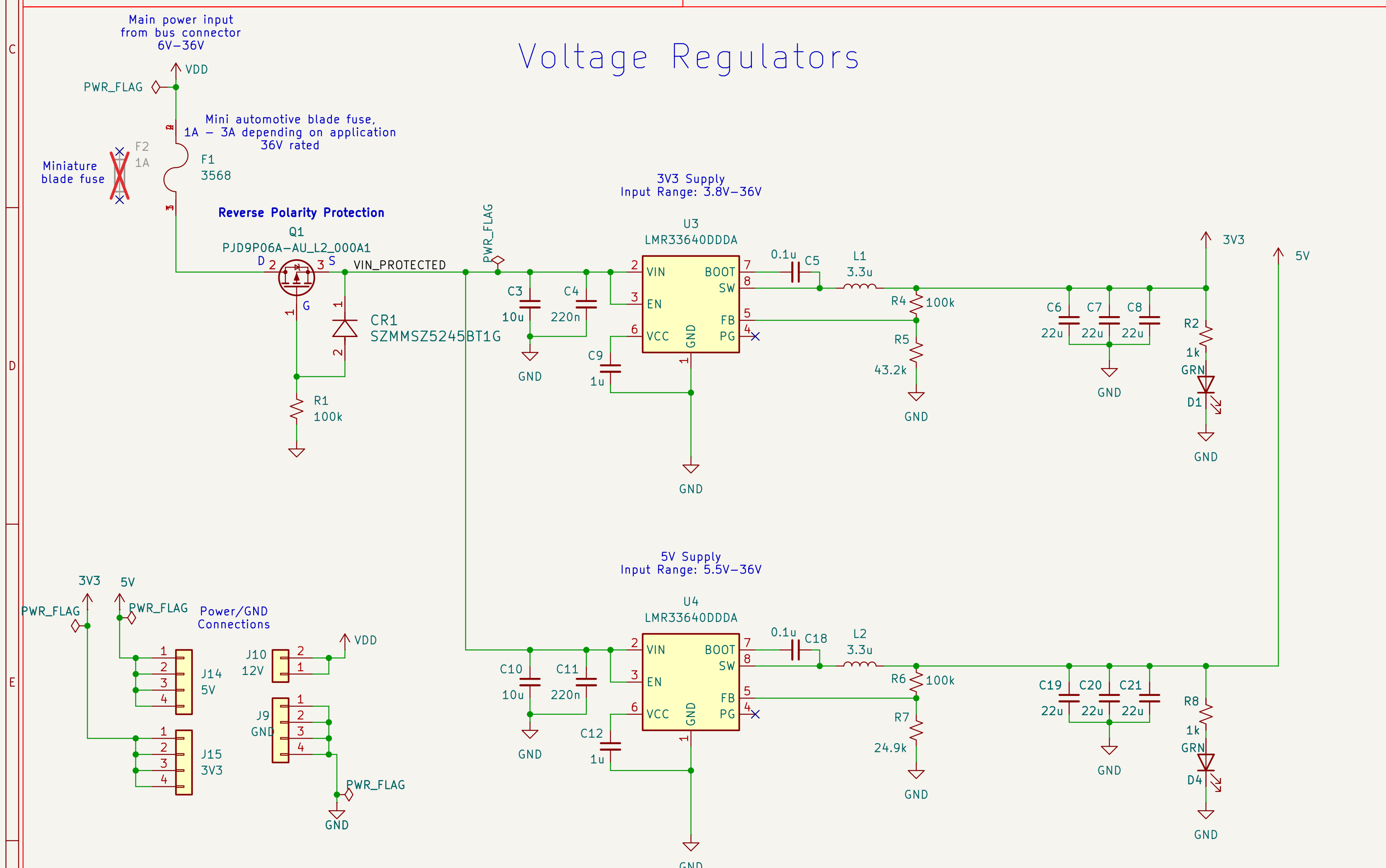
The "VMS VCU Dev Board" is the first iteration of the VCU module using a custom PCB designed to fit in a Cinch 2.25" x 3.35" enclosure

The board accepts up to a 36V DC power source (needs to be manually connected from bus connector) and contains both 3.3V and 5V regulators to power the MCU and any external sensors/components

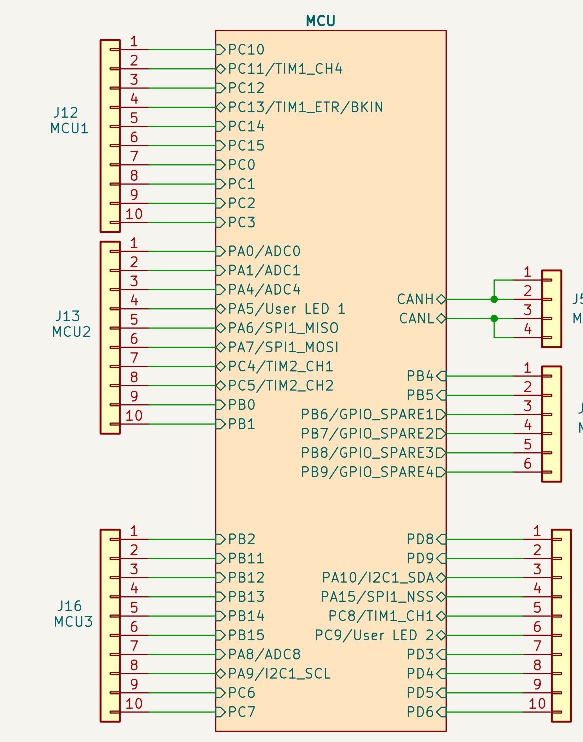
The Cinch enclosure header consists of a 12-pin connector which is to be designated as the main bus connection point and a 20-pin connector which is to be used to interface with external components and sensors

Pins from the PCB header will connect to internal headers that can be manually connected via jumpers to the appropriate source (MCU pin, 5V, 3.3V, CAN H/L, GND, etc.)

Voltage Regulators



MCU



VIKING MOTOR SPORTS (VMS)
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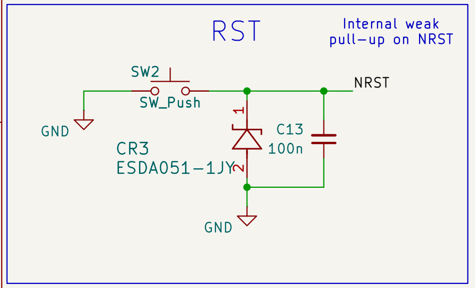
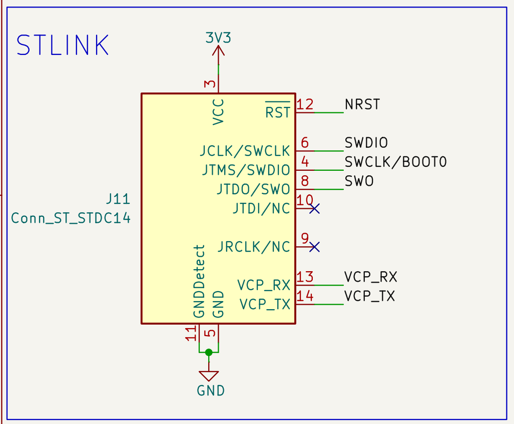
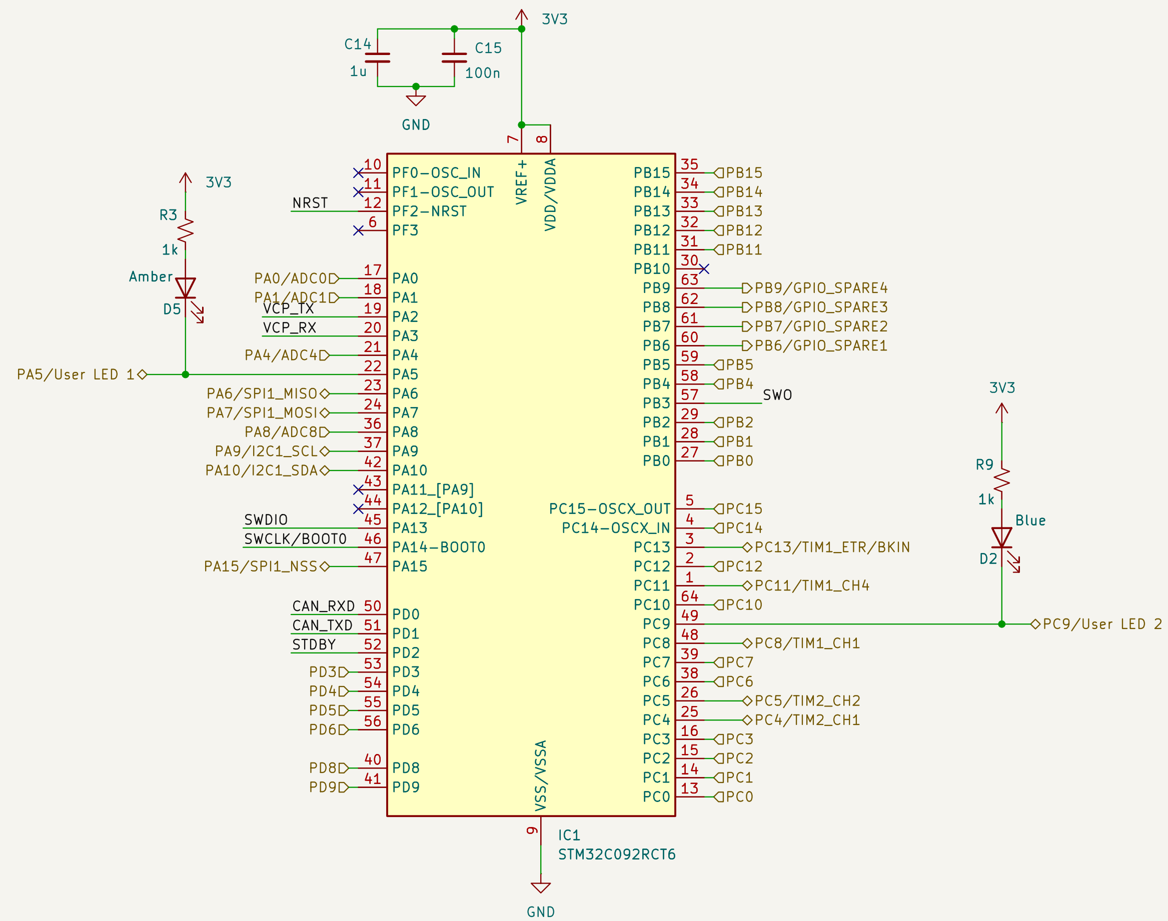
Title: VEHICLE CONTROL UNIT (VCU) Dev Board

Size: B	Date: 2026-06-23	Rev: 1.0
KiCad E.D.A. 10.0.5		Id: 1/2

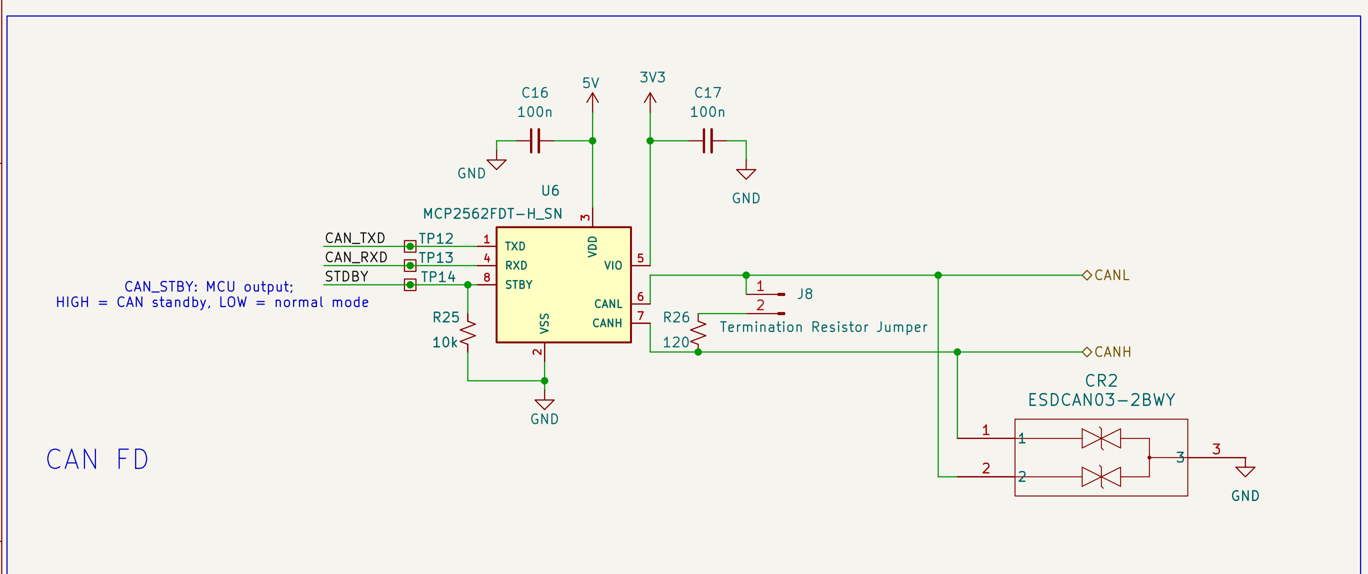






MCU PIN ASSIGNMENT / INTERFACE SUMMARY

Function	MCU Pin	Net Name	Direction / Notes
SWDIO	PA13	SWDIO	STDC14 debug / programming
SWCLK	PA14	SWCLK	STDC14 debug / programming
NRST	PF2	NRST	Hardware reset; connected to reset button and STDC14 RST
CAN RX	PD0	CAN_RX	MCU receives from CAN transceiver RXD
CAN TX	PD1	CAN_TX	MCU transmits to CAN transceiver TXD
CAN STBY	PD2	CAN_STBY	CAN transceiver standby control
GPIO Spare 1	PC1	GPIO_SPARE1	Reserved GPIO test / future expansion
GPIO Spare 2	PC2	GPIO_SPARE2	Reserved GPIO test / future expansion
GPIO Spare 3	PC3	GPIO_SPARE3	Reserved GPIO test / future expansion
Shutdown	PC0	SHUTDOWN	Shutdown output signal for system control
3.3V Supply	VDD/VDDA	+3V3	MCU digital / analog supply
Ground	VSS/VSSA	GND	MCU ground / analog reference



VIKING MOTOR SPORTS (VMS)

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Title: VEHICLE CONTROL UNIT (VCU) Dev Board

Size: B	Date: 2026-06-23	Rev: 1.0
KiCad E.D.A. 10.0.5		Id: 4/2